Math 142, F19 **Name:** Answer

Quiz 1

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (5 points) Evaluate the following indefinite integrals:

(a) $\displaystyle\int \frac{x}{x^2+2}\,dx$

Let $u = x^2+2$, then $du = 2x\,dx$

or $\frac{1}{2}du = x\,dx$

$$\int \frac{x}{x^2+2}\,dx = \int \frac{1}{u}\cdot\frac{1}{2}du$$
$$= \frac{1}{2}\ln|u| + C$$
$$= \frac{1}{2}\ln(x^2+2) + C$$

(b) $\displaystyle\int \cos\theta\sin^3\theta\,d\theta$

Let $u = \sin\theta$, then $du = \cos\theta\,d\theta$

$$\int \cos\theta\sin^3\theta\,d\theta = \int u^3\,du$$
$$= \frac{u^4}{4} + C$$
$$= \frac{\sin^4\theta}{4} + C$$

Problem 2. (5 points) Evaluate the following definite integrals:

(a) $\displaystyle\int_1^e \frac{(\ln x)^2}{x}\,dx$

Let $u = \ln x$, then $du = \frac{1}{x}dx$

when $x=1$, $u = \ln 1 = 0$

$x = e$, $u = \ln e = 1$

Thus

$$\int_1^e \frac{(\ln x)^2}{x}\,dx = \int_0^1 u^2\,du$$
$$= \frac{u^3}{3}\Big|_0^1$$
$$= \frac{1}{3} - 0 = \frac{1}{3}$$

(b) $\displaystyle\int_0^1 xe^{-x^2}\,dx$

Let $u = -x^2$, then $du = -2x\,dx$

or $-\frac{1}{2}du = x\,dx$

When $x=0$, $u = -0^2 = 0$

$x = 1$, $u = -x^2 = -1$

Thus

$$\int_0^1 xe^{-x^2}\,dx = \int_0^{-1} e^u\left(-\frac{1}{2}\right)du$$
$$= -\frac{1}{2}e^u\Big|_0^{-1}$$
$$= -\frac{1}{2}(e^{-1} - e^0)$$
$$= \frac{1}{2}\left(1 - \frac{1}{e}\right)$$